

# ECE 236 / ME 236: Nonlinear Control Systems

Winter 2026, 4 credits

|                    |                                                          |                      |                                                                                   |
|--------------------|----------------------------------------------------------|----------------------|-----------------------------------------------------------------------------------|
| <i>Instructor:</i> | Prof. Andrew R. Teel<br>teel@ucsb.edu                    | <i>Lectures:</i>     | TTh: 10:00 – 11:50 am<br>(Phelps 1437)                                            |
|                    |                                                          | <i>Office Hours:</i> | TTh: 1 – 2:30pm<br>(5119A HFH)                                                    |
| <i>Text:</i>       | H.K. Khalil<br><i>Nonlinear Control</i><br>Pearson, 2015 | <i>Grading:</i>      | Homework (3-4 assignments) 100 %<br>(Assignments may involve Matlab and Simulink) |

## COURSE TOPIC

Nonlinear stability theory and nonlinear control design.

## COURSE OUTLINE

### 1. Fundamental properties of differential equations

Existence/uniqueness, finite-escape time, limit cycles, illustrations on 2-d systems *Ch. 1-2*.

### 2. Basic stability theory

Lyapunov functions, linearization, invariance principle, converse theorems, input-to-state stability. *Ch. 3-4*.

### 3. Input-output stability

$\mathcal{L}_2$ -stability, small-gain theorem, passivity, absolute stability. *Chapters 5-7*.

### 4. Selected topics in nonlinear control design

Normal forms, state feedback stabilization, observers, output feedback *Ch. 8-13*.

## PREREQUISITES

Some knowledge of Linear Systems (ECE 230A /ME 243A) assumed.

## Course web page

On Canvas.